

STATEMENT OF WORK

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STATEMENT OF WORK (SOW)

Title: High Throughput Bioreactor

Requiring Activity Name: Walter Reed Army Institute of Research (WRAIR), Pilot Bioproduction Facility (PBF)

1. INTRODUCTION

The Walter Reed Army Institute of Research (WRAIR) provides unique research capabilities and innovative medical solutions to a range of Force Health Protection and Readiness challenges currently facing U.S. Service Members, along with threats anticipated during future operations.

Through both times of peace and war, infectious diseases have killed, sickened, and disabled far more Service Members than bombs and bullets. WRAIR has created a model of vaccine and therapeutic development that is unique, nimble, and responsive to dynamically evolving infectious disease threats of military importance. WRAIR, with its unparalleled expertise, facilities, and international network, has developed many vaccines and drugs in use today by military and civilian medicine around the globe.

In addition to protecting against infectious disease threats, WRAIR conducts basic and applied research that promotes psychological resilience, enhances neurological functioning, and improves operational readiness among U.S. Service Members. By using a continuum of research from basic science to applied field studies, WRAIR creates innovative solutions to enhance Service Members' resilience and recovery. Research in military psychiatry and neuroscience has been an integral part of WRAIR's portfolio for more than 70 years. WRAIR researchers are recognized leaders in operational medical research for the military in a number of fields, including brain trauma, blast exposure, deployment stressors, and sleep management.

2. PURPOSE AND BACKGROUND

The Pilot Bioproduction Facility (PBF) is a current Good Manufacturing Practice (cGMP) compliant, Phase I pharmaceutical manufacturing facility at WRAIR. The PBF specializes in developing vaccines and biologics for military-relevant infectious disease threats. Production is pilot-scale; 500 – 2,000 doses per finished lot of products. Products are released for use in Phase I - II clinical studies under Investigational New Drug applications (INDs). PBF support has been critical to the development efforts of the Department of Defense (DoD) in the advancement of numerous vaccines. These vaccines include those for hepatitis A, meningitis, dengue fever, malaria, adenovirus, Japanese encephalitis, shigellosis, zika, and, most recently Covid-19.

The PBF Manufacturing Science and Technology team is required to develop a Monoclonal Antibody (mAb) production capability for the development of biologics and therapeutics. A high throughput bioreactor is required for rapid development of scalable fermentation processes using multiple parallel bioreactors and a fully automated liquid handling platform, enabling researchers to test numerous conditions simultaneously, thereby accelerating the development process. This technology has been globally adopted for various bioprocess development and optimization applications.

The existing process development system at PBF, which relies on traditional bioreactors, often exhibits lower throughput, necessitating more time and resources to evaluate multiple conditions and variables. PBF requires a high throughput bioreactor with capability to fulfill the production demands for sufficient GMP material to support up to 2,000 vials of final drug product. The PBF also requires the system to support software that is 21 CFR Part 11 compliant.

The purpose of this requirement is to acquire the bioreactor system; a fully integrated high throughput system featuring easy-connect bioreactors and flexible software, enabling scientists to manage a significantly higher number of experiments simultaneously while reducing the cost per experiment. This advanced technology is optimal for scaling down processes and will facilitate efficient and seamless scalability, ensuring a direct transition to production.

3. SCOPE

The objective of this requirement is to procure a high throughput bioreactor as new technology to improve the WRAIR PBF's on-site monoclonal antibodies production capabilities. The bioreactor shall include the high throughput workstation with 12 bioreactor stations, system computer, software, system installation, External Sampling Monitor (ESM), chiller, auto cleaning-in place (CIP) module, head space gassing, qualification, and a 12-month warranty and first year service and support package, including access to a dedicated support team with comprehensive reaction time commitment and preventive maintenance, and training for two operators. In addition, this requirement shall include three one-year option periods for maintenance and service.

The equipment and associated materials shall be delivered six to nine after contract award with on-site installation, Site Acceptance Testing (SAT), qualification, and user training within 30 days of equipment receipt. Additionally, the Contractor shall include a one-year warranty for all system hardware and software.

4. PERIOD OF PERFORMANCE

The PoP for this contract is as follows:

Base period	01 June 2025 – 31 May 2026
Option Period 1	01 June 2026 – 31 May 2027
Option Period 2	01 June 2027 – 31 May 2028
Option Period 3	01 June 2028 – 31 May 2029

5. WORK SCHEDULE AND LOCATION

The delivery location is 503 Robert Grant Ave, Silver Spring, MD 20910. The Contracting Officer's Representative (COR) will coordinate with the Contractor for any WRAIR access that may be required. Scheduled on-site visits shall occur during business hours between the hours of 0800 to 1600 Eastern Standard Time (EST), Monday through Friday. Note that the final placement and securing of the equipment in the laboratory, will be performed by Medical Maintenance, with Contractor oversight due to the equipment's sensitive instrumentation and potential need to recalibrate the instruments once equipment has been secured in laboratory.

6. PERFORMANCE REQUIREMENTS

Objectives: To construct, deliver, install, test, and qualify a complete system, including all required hardware, software, tubing, cabling, and instrumentation that meets the Minimum Essential Characteristics (MECs) described in Attachment 2.

6.1 MECs Attachment 2

6.2. Tasks

6.2.1. Bioreactor Equipment and Installation.

The Contractor shall provide all hardware, software, equipment, devices, and initial consumables required for a complete, operational bioreactor system (**A.001**). This shall include the following tasks:

6.2.1.2. The Contractor shall provide a 21 CFR Part 11 assessment as part of the turnover package (**A.001**).

6.2.1.3. The Contractor shall review and confirm bioreactor and all auxiliary components with the Contracting Officer (CO) and COR prior to starting design and/or construction.

6.2.1.4. The Contractor shall coordinate the delivery and installation of the complete system.

6.2.1.5. The Contractor shall remove all packaging material, used or damaged consumables, and trash generated during the shipping, installation, and testing of the equipment.

6.2.1.6. The Contractor shall complete a contractor qualification questionnaire (**A.001**).

6.2.3. Site Acceptance Testing (SAT)

6.2.3.1. The Contractor shall provide all documentation required to complete a SAT (**A.002**).

6.2.3.2. The Contractor shall provide on-site support for the execution of the SAT.

6.2.3.3. The Contractor shall coordinate all execution activities with the COR, CO, or designated representative and the execution will be 100% witnessed by the government representative.

6.2.4. Installation / Operation Qualification

6.2.4.1 The Contractor shall provide a complete Installation and Operational Qualification (IQQ) documentation package required to complete the qualification testing (**A.003**).

6.2.4.2 The Contractor shall provide on-site support for the execution of the IQQ.

6.2.4.3 The Contractor shall coordinate all IQQ execution activities with the CO or designated representative. IQQ execution will be 100% witnessed and approved by the CO, COR, or designated representative.

6.2.5 Training. The Contractor shall provide end-user, administrative, and maintenance training for a minimum of two individuals. The training plan shall be provided to the CO, COR, or designated representative a minimum of 30 days prior to training. User training shall be completed within 30 days of equipment receipt (**A.004**)

6.2.6 Annual Maintenance and System Calibration/Support Services. The contractor shall provide annual preventative maintenance and system calibration service, as described below, to assure optimal performance of the system.

6.2.6.1. On-site Repairs:

6.2.6.1.1. Contracted services are to be performed throughout the period of performance of this contract. The Contractor is required to respond within 48 hours.

6.2.6.1.2. Prior to reporting to a service call, the Contractor's service representative shall notify medical maintenance via email (usarmy.detrick.medcom-wrair.mbx.medical-maintenance@health.mil) and shall report in person or telephonically notify the maintenance manager, with building and room number as well as telephone number, prior to commencing services during normal duty operating hours. During other than normal operating hours, the contractor's service representative shall report to the Administrative Officer of the Day (AOD) and notify medical maintenance via email (**A.005**).

6.2.6.1.3. The Government and the Contractor's service representative shall exchange hazard communication information before the commencement of any repair.

6.2.6.1.4. When required, the Contractor's service representative shall comply with the Occupational Safety and Health Administration (OSHA) lockout/tag-out standards while performing maintenance on Medical Devices (MDs)/Medical Device Systems (MDSs).

6.2.6.1.5. Upon completion of services by the Contractor's service representative, a service report shall be provided to the COR or the AOD and submit an electronic copy of the service report to medical maintenance at usarmy.detrick.medcom-wrair.mbx.medical-maintenance@health.mil. The service report shall provide detailed information regarding the cause of the MD/MDS malfunction and corrective action taken. It shall include, at a minimum, the time required to complete the work, price of labor (hourly rate), and a list of parts replaced with part numbers and prices for each part prior to leaving the facility. **(A.006)**

6.2.6.1.6. In the event all information is not available to the Contractor's service representative when services are performed, the initial service report/invoice shall include all available information to close work order within five days. The Contractor shall provide the balance of the required information to the clinical engineering manager no later than ten days after services are completed. **(A.006)**

6.2.6.1.7. The Contractor's service representative shall be factory trained and have a minimum of two years of experience working on the contracted MD/MDS. In addition, services shall be performed by a Field Service Engineer (FSE) that is certified by the manufacturer. No other FSEs are authorized to work on these instruments.

6.2.6.1.8. The Contractor shall furnish all software updates/patches issued by the MD/MDS manufacturer.

6.2.6.1.9. The Contractor shall have access to all necessary diagnostic software (if applicable). The Contractor shall use only OEM new or OEM refurbished repair parts.

6.2.6.1.10. For all contracts that require calibration of systems, the Contractor shall complete DD Form 2163 or DD Form 2164 (Verification/Certification Worksheet) in accordance with the instructions provided in AR 40-61. Additionally, continuation sheet shall be attached to the DD Form 2164 indicating the manufacturer, model, serial number, and date of calibration expiration of all items of test, measurement, and diagnostic equipment (TMDE) used to perform the calibration. Submit an electronic copy of the completed Verification/Certification Worksheet to medical maintenance at usarmy.detrick.medcom-wrair.mbx.medical-maintenance@health.mil and provide a copy to the COR within five days of work completion. **(A.007)**

7. DELIVERABLES

The contractor shall complete and submit deliverables in accordance with the "Deliverables Table" below. Unless specified in the SOW within the description of the deliverable, all deliverables will be made to the COR electronically.

Number	SOW Ref.	Title/Description	Due
A.001	6.2.1.1, 6.2.1.2, 6.2.1.6.	Bioreactor System Equipment and Documentation (including 21 CFR Part 11 Assessment and Qualification Questionnaire)	Due within 6 months and no later than 9 months (based on the duration to manufacture and delivery time.
A.002	6.2.3.1.	Site Acceptance Testing (SAT) Documentation	Due within 30 days of equipment receipt date
A.003	6.2.4.1.	Installation and Operational Qualification (IQOQ)	Due within 30 days of equipment receipt date
A.004	6.2.5.	Training Plan	Due 30 days prior to training, with training completed within 30 days of equipment receipt date
A.005	6.2.6.1.2.	Medical Maintenance Notification	Due prior to reporting to a service/repair call.
A.006	6.2.6.1.5., 6.2.6.1.6.	Service Report/Preventative Maintenance Report	Initial service report due prior to leaving facility or no later than

			five days after service is completed if additional information is needed; final report no later than 10 days after service is completed.
A.007	6.2.6.1.10.	Calibration of System(s)	Send electronic copy to Medical Maintenance and COR within five days of work completion.
A.008	8.1.5.	OSHA 300A	Due by February 1 of each year
A.009	9.3.2.	TARP Training Report	Due within 30 days after contract award, annually thereafter

7.2. Inspection and Acceptance

The COR is delegated with oversight of specific technical and functional responsibilities to include inspection and acceptance of all products, incoming shipments, documents, and services.

Certification by the Government of satisfactory performance provided is contingent upon the Contractor adhering to the performance standards contained in the SOW and all terms and conditions of this order, including all modifications.

7.3. COR Information

TBD
WRAIR, PBF
501 Robert Grant Rd
Silver Spring, MD 20910

7.4. Contract Management

7.4.1. Government Management. The COR for this contract will be appointed by the CO. The COR will serve as the primary point of contact for all activities and issues that occur under this contract. Only the CO has the authority to enact changes to this award.

7.4.2 Contractor Organization. The Contractor shall establish clear organizational lines of authority and responsibility to ensure effective management of the resources assigned to the requirement. The Contractor shall ensure that all Contractor personnel are advised of their chain of command and who they should contact if they have questions.

8. COMPLIANCE DOCUMENTS

8.1. General Compliance

8.1.1. Federal Law. The services provided under this contract must be conducted in accordance with all Federal, DoD, Department of Health Agency (DHA), and United States Army Medical Research and Development Command (USAMRDC) laws command laws, regulations, policies, and/or procedures that govern the conduct of regulated research.

8.1.2. Applicable Publications

- The Army Physical Security Program (AR 190-13)
- U.S. Army Corps of Engineers Safety and Health Requirements Manual (EM 385-1-1)
- Medical Logistics Policies (AR 40-61)
- Antiterrorism Force Protection: Security of Personnel, Information, and Critical Resources (AR 525-13)

- Operations Security (AR 530-1)
- Insider Threat Program (DHA-AI 097)
- Testing and Calibration Laboratories (International Organization for Standardization (ISO) 17025)
- 21 CFR Part 11

8.1.3. Personally Identifiable Information (PII) and Protected Health Information (PHI)

The contractor shall comply with the DHA Procedures, Guides, and Information (PGI) 224 – Protection of Privacy and Freedom of Information, revised 17 May 2015. The contractor may access PII in the course of task response to the requirements of this contract, as such the contractor shall comply with the document entitled “Personally Identifiable Information (PII), Protected Health Information (PHI) and Federal Information Requirements”.

The Contractor shall establish appropriate administrative, technical, and physical safeguards to protect any and all Government data. The Contractor shall also ensure the confidentiality, integrity, and availability of Government data in compliance with all applicable laws and regulations, including data breach reporting and response requirements, in accordance with DFAR Subpart 224.1 (Protection of Individual Privacy), which incorporates by reference DoDD 5400.11, “DoD Privacy Program,” May 8, 2007, and DoD 5400.11-R, “DoD Privacy Program,” May 14, 2007. The contractor shall also comply with federal laws relating to freedom of information and records management. The Contractor shall comply with all requirements of the Health Insurance Portability and Accountability Act of 1996 (HIPAA) (Pub. L. 104-191), as implemented by the HIPAA Privacy and Security Rules codified at 45 CFR Parts 160 and 164, and as further implemented within the Military Health System (MHS) by DoD 6025.18-R, “DoD Health Information Privacy Regulation,” January 24, 2003, and DoD 8580.02-R, “DoD Health Information Security Regulation, July 12, 2007.

IT services that collect, maintain, use, or disseminate personally identifiable information must be managed in a manner that protects privacy, in accordance with section 552a of Title 5, U.S.C. (Reference (aa)), DoD 5400.11 (Reference (ab)) and DoD 5400.11-R (Reference (ac)). DoDI 5400.16 (Reference (ad)) established the guidance for development, review, and approval of Privacy Impact Assessments, in accordance with chapter 36 of Title 44, U.S.C. (Reference (ae)).

8.1.4. Information Technology (IT) Compliance

8.1.4.1. Software. The Contractor shall permit the Government to perform software vulnerability scans on acquired software to identify potential weaknesses in the code. Collaboration between the Contractor and the Government is required to address any vulnerabilities detected during these scans. Additionally, the Contractor is responsible for providing ongoing support to patch vulnerabilities throughout the software’s lifespan. All Commercial-Off-The-Shelf (COTS) software must be registered on SAM.gov and have a cage and Data Universal Number System (DUNS) number.

8.1.4.2. Hardware. Contractors are required to offer IT maintenance warranty support for any purchased hardware until hardware reaches its End of Life (EOL). Furthermore, Contractors must also provide patching support to include firmware and driver updates for any vulnerabilities uncovered during Government vulnerability scanning. All hardware must be Trade Agreement Act (TAA) compliant and all computers must be Trusted Platform Module (TPM) 2.0 compliant.

8.1.4.3. IT Systems. Contractors should commit to work with and assist the Government agency (purchaser) to obtain an approval to operate on or with the DHA environment. This approval to operate must come from the DHA Authorizing Official (AO) using National Institute of Standards and Technology (NIST) 800-37 and NIST 800-53 controls to obtain approval from DHA AO before the hardware and/or software can be used with the DHA environment on the DHA network or as a standalone in a research lab.

8.1.4.4. Mitigation of Vulnerabilities.

8.1.4.4.1. Maintain all equipment/product versions provided pursuant to the construction or other procurement contract, to include the supported operating system, by issuing patches/updates to mitigate vulnerabilities [e.g., Information Assurance Vulnerability Alert (IAVA) or Information Assurance Vulnerability Bulletins (IAVB)] and

network security configurations [e.g., Defense Information System Agency (DISA) Security Technical Implementation Guides (STIGs)]. The contractor shall adhere to DHA guidance for remediation or mitigation of vulnerabilities associated with the equipment/product.

8.1.4.4.2. Test all IAVAs, patches, updates, and upgrades in the test/laboratory environment before implementing in a production environment. All production environment IAVAs, patches, updates, and upgrades shall be approved by the Medical Treatment Facility (MTF's) Change Control Board prior to contractor implementation. Approved changes shall be implemented on a quarterly basis.

8.1.5. Safety Laws and Regulations. The contractor shall maintain a safe workplace and comply with occupational health and safety rules and regulations, to include OSHA number 1904.32, and maintain an OSHA 300A log for all reportables. The contractor shall provide a quarterly report of all mishaps, regardless of class or reportability. In addition, the contractor shall deliver the annual OSHA 300A report by February 1 of each year (A.008).

9. OTHER TERMS, CONDITIONS, AND PROVISIONS

9.1. Invoices

The contractor shall submit invoices monthly and/or annually to Wide Area Work Flow (WAWF) with sufficient documentation to provide certification of invoice.

The contractor is responsible for properly preparing and forwarding to the appropriate Government official, the invoice and receiving report for payment. Invoices submitted for payment, which do not contain Contract Line Item Number (CLIN) and the Accounting Classification Reference Number (ACRN) information, will be rejected. The Wide Area Workflow – Receipt and Acceptance application is the required method of submission.

9.2. Access to Government Facilities

9.2.1. Contractors That Do Not Require a Common Access Card (CAC), But Require Access To A DoD Facility or Installation: The contractor and all associated subcontractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB); applicable installation, facility, and area commander installation/facility access; and local security policies and procedures (provided by Government representative).

9.2.2 Non-Escort/Escorted Badge: Non-escort/escorted badges are completed by the COR on a Memorandum to the Operations and Security (S2) office, signed by the S2 shop, and provided to the Staff Duty desk for said individuals. Prior to receiving the non-escort/escorted badge, individuals will be vetted by the local Program Management Office (PMO). Vetting is done by completing the Fort Detrick Form 190-13, filled out by the COR with the receiving person's information and signed by the S2 office. Once signed by the S2 office, the information is then passed to the Visitor Control Center (VCC) for background checks, which must be accomplished two weeks ahead of scheduled work. All persons entering the Forrest Glenn Annex shall have a background check conducted, whether local or through other official means. Visitors will be escorted at all times in WRAIR facilities.

9.3. Training: Anti-Terrorism and Operational Security

9.3.1. Threat Awareness Reporting Program (TARP). Per Defense Health Agency-Administrative Instruction (DHA-AI) 097, Insider Threat Program, contractor employees with security clearances must receive annual Insider Threat Awareness training within 30 days of initial employment, and refresher training annually either through instructor-led or computer-based (A.009).

10. ATTACHMENTS

Attachment 1 – Acronyms List

Attachment 2 - MECs

ATTACHMENT 1 – ACRONYMS LIST

Acronym	Definition
ACRN	Accounting Classification Reference Number
AO	Authorizing Official
AOD	Administrative Officer of the Day
AR	Army Regulation
ATO	Antiterrorism Officer
AT OPSEC	Antiterrorism and Operational Security
CAC	Common Access Card
CFR	Code of Federal Regulations
cGMP	Current Good Manufacturing Practices
CIP	Clean In Place
CLIN	Contract Line-Item Number
CO	Contracting Officer
COR	Contracting Officer Representative
COTS	Commercial-Off-The-Shelf
DA	Department of the Army
DFARS	Defense Federal Acquisition Regulation Supplement
DHA	Department of Health Agency
DISA	Defense Information System Agency
DoD	Department of Defense
DoDI	Department of Defense Instruction
DUNS	Data Universal Number System
EIT	Electronic and Information Technology
EOL	End of Life
ESM	External Sampling Monitor
EST	Eastern Standard Time
FAR	Federal Acquisition Regulation
FSE	Field Service Engineer
GMP	Good Manufacturing Practice
HIPAA	Health Insurance Portability and Accountability Act
HT	High Throughput
IA	Information Assurance
IAVA	Information Assurance Vulnerability Alert
IAVB	Information Assurance Vulnerability Bulletins
ID	Identification
IEC/EN	International Electrotechnical Commission
IND	Investigational New Drug applications
IQOQ	Installation and Operational Qualification
ISO	International Organization for Standardization
IT	Information Technology
mAB	Monoclonal Antibody

MD	Medical Devices
MDS	Medical Device Systems
MECs	Minimum Essential Characteristics
MHS	Military Health Systems
MTF	Medical Treatment Facility
NAC	National Agency Check
NACI	National Agency Check with Inquiries
NCIC-III	National Crime Information Center Interstate Identification Index
NIST	National Institute of Standards and Technology
OEM	Original Equipment Manufacturer
OPSEC	Operational Security
OSHA	Occupational Safety and Health Administration
PBF	Pilot Bioproduction Facility
PC	Personal Computer
PGI	Procedures, Guides, and Information
PHI	Protected Health Information
PII	Personal Identification Information
PMO	Project Management Office
PoP	Period of Performance
SAT	Site Assessment Testing
STIGs	Security Technical Implementation Guides
SOP	Standing Operating Procedure
SOW	Statement of Work
TAA	Trade Agreement Act
TARP	Threat Awareness Reporting Program
TMDE	Test, Measurement, and Diagnostic Equipment
TPM	Trusted Platform Module
TSDB	Terrorist Screening Database
USAMRDC	United States Army Medical Research and Development Command
U.S.C.	United States Code
UIC	Unit Identification Code
US	United States
WAWF	Wide Area Work Flow
WRAIR	Walter Reed Army Institute of Research

ATTACHMENT 2 - Minimum Essential Characteristics:

High Throughput Bioreactor

The bioreactor shall meet the following MECs:

- Processes from up to 12 parallel, simultaneous cultures that can be used for support Design of Experiments (DOE).
- Allows for the culture of both mammalian and microbial cells using available single use vessels with appropriate dual 20 mm pitched-blade or Rushton impeller with agitation speed ranging from 100 – 4500 rpm
- Provides tuning options including programmable Proportional, Integral, and Derivative (PID) control loops.
- Provides full control for pH and dissolved oxygen (DO) inclusive of control of gas to headspace and sparge and feeds.
- Provides an integrated sterile operating environment that allows for fully automated aseptic processing from batching to feeds and sample management with limited operator interaction.
- Allows connection to analytical instruments such as cell counter(s) and chemical/gas analyzers for automated sample processing and disposable DO, CO₂ and pH probes with automated feed and feedback controls and option for connection to exhaust gas analysis for Oxygen Uptake Rates (OUR)/Carbon dioxide Evolution Rates (CER)
- Allows for robotic liquid handler to sample from each bioreactor in volumes between 50 uL and 10 mL.
- Allows for robotic liquid handler to make liquid additions to bioreactors between 30 uL and 10 mL.
- Incorporates an automated pipette rinse with decontamination liquid.
- Allows for self-contained chiller for bioreactor cooling.
- Displays real-time data, store, and provide data in an off-line format and provide capability for external monitoring; software application enables predictive scale conversion.
- Provides for small volume high throughput with scalable features at a 100 - 250 mL working volume.
- Allows for scalability up to 30 L
- The system software and design shall be consistent with 21 CFR Part 11 requirements.
- Ensures the workstation (laptop) will keep working on battery and the controller to switch off in the event of power failure and without an external Uninterrupted Power Supply (UPS), with an option to connect to an external UPS to ensure an uninterrupted process.

Bioreactors and Process Control

- Vessel design shall be geometrically consistent with standard production scale bioreactors and fermenters and shall allow for scalable pitched blade (bioreactor) or Rushton (fermenter) impellers.
- Provides individual bioreactor controls including:
 - Individual temperature control with heating and cooling
 - Individual impeller speed control
 - Individual gas flow control

- Individual pH testing and control
 - Individual sampling
- Allows for headspace gassing. Bioreactors shall be capable of simultaneous controlled addition of up to three gasses with mass flow sensor, either as sparge or overlay:
 - Mammalian
 - O₂
 - CO₂
 - N₂ | air
 - Microbial
 - O₂
 - Air
 - N₂
- Capable of monitoring dissolved O₂ via an autoclavable or single-use probe.
- Capable of monitoring pCO₂ range from 15-200 mmHg
- Capable of monitoring DO range from 0-200%
- Capable of monitoring pH over the range of the process (2 – 8.5).
- Automatically monitors and controls the media temperature with an operational control range of 32-37°C and temperature probe shall be capable of monitoring between 0 - 55 °C.
- Allows for total gas or air flow up to 550 mL/min.
- Provides four positive displacement liquid pumps per bioreactor for high precision feed additions at low flow rates.
- Allows for integrated clean in place (CIP) | sterilize in place (SIP) for pumps and liquid lines.